

Supplementary Material

for

Projection-based Riemannian federated learning with partial participation

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S1. Preliminaries

Lemma S1.1 (Lemma A.1 in [1]). *For all $x \in \mathcal{M}$ and $u \in \mathbb{R}^{d \times p}$ such that $\|u\| \leq \gamma$,*

$$\|\mathcal{P}_{\mathcal{M}}(x + u) - x\| \leq 2\|u\|. \quad (\text{S1})$$

The constraint $\|u\| \leq \gamma$ ensures that $x + u$ remains within the tubular neighborhood where $\mathcal{P}_{\mathcal{M}}$ is well-behaved. Equation (S1) introduces a benign metric distortion factor that upper-bounds the displacement induced by a single retraction.

Lemma S1.2. *Let $\mathcal{M}$ be 2γ -proximally smooth.*

1. *Lipschitz continuity of tangent projectors: There exists $L_{\text{dP}} > 0$ such that for all $x, y \in \mathcal{M}$ with $\|x - y\| \leq \gamma$,*

$$\|P_x - P_y\| \leq L_{\text{dP}}\|x - y\|. \quad (\text{S2})$$

2. *Quadratic remainder: For all $x \in \mathcal{M}$ and u with $\|u\| \leq \gamma$,*

$$\|\mathcal{P}_{\mathcal{M}}(x + u) - (x + P_x u)\| \leq O(\|u\|^2).$$

The first point follows from [2, Theorem 4.8], stating that proximal smoothness implies $\mathcal{P}_{\mathcal{M}}$ is $\mathcal{C}^{1,1}$ on $U_{\mathcal{M}}(\gamma)$, so its differential $x \mapsto P_x$ is Lipschitz. The second point is [3, Lemma 4.3] and shows that retracting $x + u$ differs from the first-order approximation $x + P_x u$ by a purely quadratic term. In the FL setting, it also explains why different server aggregations (retract–lift vs. projection of the mean) diverge only in higher-order terms.

Lemma S1.3 (Lemma 4.2 in [3]). *Let $f : \mathcal{M} \rightarrow \mathbb{R}$ be such that f admits a C^1 extension to $U_{\mathcal{M}}(\gamma)$ with Euclidean gradient ℓ -Lipschitz on $U_{\mathcal{M}}(2\gamma)$, i.e.*

$$\|\nabla f(x) - \nabla f(y)\| \leq \ell \|x - y\| \quad x, y \in U_{\mathcal{M}}(2\gamma).$$

Let $G := \max_{x \in \mathcal{M}} \|\nabla f(x)\| < \infty$. For all $x, y \in \mathcal{M}$,

$$f(y) \leq f(x) + \langle \text{grad } f(x), y - x \rangle + \frac{L_{\mathcal{M}}}{2} \|y - x\|^2, \quad L_{\mathcal{M}} := \ell + \frac{G}{2\gamma}. \quad (\text{S3})$$

Remark S1.4. The constant $L_{\mathcal{M}} = \ell + G/(2\gamma)$ combines ambient smoothness ℓ with a curvature correction $G/(2\gamma)$ arising from the normal component of $y - x$; see [3]. For large γ (flat geometry), $L_{\mathcal{M}} \approx \ell$.

In the sequel, under our assumptions, since F is $\bar{\ell}$ -smooth, we have by the previous Lemma

$$F(y) \leq F(x) + \langle \text{grad } F(x), y - x \rangle + \frac{L_{\mathcal{M}}}{2} \|y - x\|^2, \quad L_{\mathcal{M}} = \bar{\ell} + \frac{G}{2\gamma}.$$

Client objectives may differ both in curvature (Lipschitz constants) and in gradient directions, reflecting data heterogeneity across devices. Our notation keeps these two sources of heterogeneity separate: ℓ_i describes smoothness of client i , whereas the dispersion of $\text{grad } f_i$ around $\text{grad } F$ is captured by the variance parameter σ^2 in Assumption 4.5 of the main paper. This separation clarifies which terms arise from curvature versus statistical variability.

In Algorithm 1 of the main paper, randomness arises from two sources: (i) the client subsampling S_t , and (ii) the data (minibatch) sampling on each selected client $i \in S_t$ and local step j . Let $(\mathcal{F}_t)_{t \geq 0}$ be the natural filtration generated by $\{x_0, S_0, \mathcal{B}_{0,\cdot}^{(\cdot)}, \dots, S_{t-1}, \mathcal{B}_{t-1,\cdot}^{(\cdot)}\}$, so that x_t is $\mathcal{F}_t$ -measurable. We write $\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_t]$.

Conditional on $\mathcal{F}_t$, the randomness in the current round t comes from S_t and from the new minibatches. Under Assumption 4.5 of the main paper, standard sampling without replacement computations (see e.g. [4]) give

$$\begin{aligned} \mathbb{E}_t \left[\frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) \right] &= \text{grad } F(x_t), \\ \mathbb{E}_t \left[\left\| \frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) - \text{grad } F(x_t) \right\|^2 \right] &\leq \frac{n-k}{k(n-1)} \sigma^2. \end{aligned} \quad (\text{S4})$$

Similarly, for $\bar{\ell}_{S_t} := \frac{1}{k} \sum_{i \in S_t} \ell_i$, we have

$$\mathbb{E}_t[\bar{\ell}_{S_t}] = \bar{\ell}, \quad \mathbb{E}_t[\bar{\ell}_{S_t}^2] = \bar{\ell}^2 + O(1). \quad (\text{S5})$$

S2. Proof of Lemma 4.7 (Tube-stability and linearization)

Fix round t and client $i \in S_t$. Using the definition of the projection-based retraction, the local update is

$$x_{t,j+1}^{(i)} = R_{x_{t,j}^{(i)}} \left(-\eta \widehat{\text{grad}} f_i(x_{t,j}^{(i)}) \right) = \mathcal{P}_{\mathcal{M}}(x_{t,j}^{(i)} - \eta \widehat{\text{grad}} f_i(x_{t,j}^{(i)})).$$

(a) Path length. By induction on j . The base case $j = 0$ is trivial. For the inductive step, let $u := -\eta \widehat{\text{grad}} f_i(x_{t,k}^{(i)})$. Then $\|u\| \leq \eta G \leq \gamma/4 < \gamma$, so (S1) gives $\|x_{t,k+1}^{(i)} - x_{t,k}^{(i)}\| \leq 2\|u\| \leq 2\eta G$. By triangle inequality:

$$\|x_{t,k+1}^{(i)} - x_t\| \leq \|x_{t,k+1}^{(i)} - x_{t,k}^{(i)}\| + \|x_{t,k}^{(i)} - x_t\| \leq 2\eta G + 2G\eta k = 2G\eta(k+1).$$

At $k = \tau - 1$: $\|x_{t,\tau}^{(i)} - x_t\| \leq 2G\alpha \leq \gamma/2$.

(b) Feasibility. For $y_{t,j}^{(i)} := x_{t,j}^{(i)} - \eta \widehat{\text{grad}} f_i(x_{t,j}^{(i)})$, triangle inequality and part (a) give

$$\|y_{t,j}^{(i)} - x_t\| \leq \|x_{t,j}^{(i)} - x_t\| + \eta G \leq 2G\alpha + \eta G \leq \frac{\gamma}{2} + \frac{\gamma}{4} < \gamma.$$

Thus $y_{t,j}^{(i)} \in U_{\mathcal{M}}(\gamma)$ and $\mathcal{P}_{\mathcal{M}}(y_{t,j}^{(i)})$ is unique.

(c) Linearization. Since $\widehat{\text{grad}} f_i(x_{t,j}^{(i)}) \in T_{x_{t,j}^{(i)}} \mathcal{M}$ and $\eta G \leq \gamma$, Lemma S1.2 yields

$$x_{t,j+1}^{(i)} = x_{t,j}^{(i)} - \eta \widehat{\text{grad}} f_i(x_{t,j}^{(i)}) + O(\eta^2).$$

Let $\zeta_{t,j}^{(i)} := \widehat{\text{grad}} f_i(x_{t,j}^{(i)}) - \text{grad} f_i(x_{t,j}^{(i)})$. We have $\mathbb{E}_t[\zeta_{t,j}^{(i)}] = 0$ and $\mathbb{E}_t\|\zeta_{t,j}^{(i)}\|^2 \leq \frac{\sigma_{\text{sg}}^2}{B}$. We can write the decomposition

$$\widehat{\text{grad}} f_i(x_{t,j}^{(i)}) = \text{grad} f_i(x_t) + \left(\text{grad} f_i(x_{t,j}^{(i)}) - \text{grad} f_i(x_t) \right) + \zeta_{t,j}^{(i)}$$

Summing over $j = 0, \dots, \tau - 1$ and using $\eta^2\tau \leq \alpha^2$ (since $\tau \geq 1$):

$$\begin{aligned}
x_{t,\tau}^{(i)} - x_t &= -\eta \sum_{j=0}^{\tau-1} \widehat{\text{grad}} f_i(x_{t,j}^{(i)}) + O(\alpha^2) \\
&= -\eta \sum_{j=0}^{\tau-1} \text{grad} f_i(x_t) - \eta \sum_{j=0}^{\tau-1} \zeta_{t,j}^{(i)} - \eta \sum_{j=0}^{\tau-1} (\text{grad} f_i(x_{t,j}^{(i)}) - \text{grad} f_i(x_t)) + O(\alpha^2) \\
&= -\eta \sum_{j=0}^{\tau-1} \text{grad} f_i(x_t) - \eta \sum_{j=0}^{\tau-1} \zeta_{t,j}^{(i)} + O(\alpha^2(1 + \ell_i))
\end{aligned}$$

Indeed,

$$\begin{aligned}
\|\text{grad} f_i(x_{t,j}^{(i)}) - \text{grad} f_i(x_t)\| &\leq \underbrace{\|P_{x_t}\|}_{=1} \|\nabla f_i(x_{t,j}^{(i)}) - \nabla f_i(x_t)\| + \|P_{x_{t,j}^{(i)}} - P_{x_t}\| \underbrace{\|\nabla \tilde{f}_i(x_t)\|}_{\leq G} \\
&\leq \ell_i \|x_{t,j}^{(i)} - x_t\| + L_{\text{dP}} G \|x_{t,j}^{(i)} - x_t\|
\end{aligned}$$

using Assumption 4.3 of the main paper and (S2). Finally,

$$\begin{aligned}
\eta \sum_{j=0}^{\tau-1} \left\| \text{grad} f_i(x_{t,j}^{(i)}) - \text{grad} f_i(x_t) \right\| &\leq \eta \sum_{j=0}^{\tau-1} (\ell_i + L_{\text{dP}} G) \|x_{t,j}^{(i)} - x_t\| \\
&\leq \eta \sum_{j=0}^{\tau-1} (\ell_i + L_{\text{dP}} G) (2G\eta j) = O(\alpha^2(1 + \ell_i))
\end{aligned}$$

using part (a).

(d) Aggregation feasibility. By part (a) and triangle inequality, $\|s_t - x_t\| \leq \frac{1}{k} \sum_{i \in S_t} \|x_{t,\tau}^{(i)} - x_t\| \leq 2G\alpha \leq \gamma/2$. Similarly, $\|u_t\| \leq 2G\alpha$ since $\|P_{x_t}\| = 1$. Both points lie in $\overline{U_{\mathcal{M}}(\gamma)}$, ensuring unique projections.

S3. Proof of Lemma 4.8 (Aggregation bias)

Recall that, given global iterate x_t at round t , the two aggregation strategies RFEDRL and RFedProj provide the next global iterate in the following way

$$x_{t+1} = \begin{cases} \mathcal{P}_{\mathcal{M}}(x_t + u_t), & u_t := \frac{1}{k} \sum_{i \in S_t} P_{x_t}(x_{t,\tau}^{(i)} - x_t) & \text{(RFedRL)}, \\ \mathcal{P}_{\mathcal{M}}(s_t), & s_t := \frac{1}{k} \sum_{i \in S_t} x_{t,\tau}^{(i)} & \text{(RFedProj)}. \end{cases}$$

Recall that from Lemma 4.7(c) in the main paper, we have

$$x_{t,\tau}^{(i)} - x_t = -\eta \sum_{j=0}^{\tau-1} \text{grad } f_i(x_t) - \eta \sum_{j=0}^{\tau-1} \zeta_{t,j}^{(i)} + O(\alpha^2(1 + \ell_i)).$$

Lifting to $T_{x_t}\mathcal{M}$ and averaging:

$$u_t = -\eta \sum_{j=0}^{\tau-1} \frac{1}{k} \sum_{i \in S_t} P_{x_t}(\text{grad } f_i(x_t)) + \xi_t + O(\alpha^2(1 + \bar{\ell}_{S_t})), \quad \xi_t := -\frac{\eta}{k} \sum_{i \in S_t} \sum_{j=0}^{\tau-1} P_{x_t} \zeta_{t,j}^{(i)}$$

By linearity of P_{x_t} , we have the key identity

$$P_{x_t}(s_t - x_t) = P_{x_t} \left(\frac{1}{k} \sum_{i \in S_t} x_{t,\tau}^{(i)} - x_t \right) = \frac{1}{k} \sum_{i \in S_t} P_{x_t}(x_{t,\tau}^{(i)} - x_t) = u_t. \quad (\text{S6})$$

The quadratic remainder decomposition (Lemma S1.2) then yields, for the two aggregation strategies:

$$x_{t+1}^{\text{RFedProj}} - x_t = \mathcal{P}_{\mathcal{M}}(s_t) - x_t = P_{x_t}(s_t - x_t) + O(\|s_t - x_t\|^2), \quad (\text{S7})$$

$$x_{t+1}^{\text{RFedRL}} - x_t = \mathcal{P}_{\mathcal{M}}(x_t + u_t) - x_t = u_t + O(\|u_t\|^2). \quad (\text{S8})$$

Combining (S6), (S7) and (S8), and using the inequalities from Lemma 4.7(d) in the main paper (namely $\|s_t - x_t\| \leq 2G\alpha$ and $\|u_t\| \leq 2G\alpha$), we obtain that for both aggregation strategies,

$$x_{t+1} - x_t = u_t + O(\alpha^2) = -\alpha \frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) + \underbrace{O(\alpha^2(1 + \bar{\ell}_{S_t}))}_{=: \bar{b}_t} + \xi_t.$$

Using (S5), we obtain $\mathbb{E}_t[\|\bar{b}_t\|] = O(\alpha^2)$ and $\mathbb{E}_t[\|\bar{b}_t\|^2] = O(\alpha^4)$. On the other hand, $\mathbb{E}_t[\xi_t] = 0$ and $\mathbb{E}_t\|\xi_t\|^2 \leq \frac{\alpha^2}{kB} \sigma_{\text{sg}}^2$.

S4. Proof of Lemma 4.9 (One-round expected descent)

Since $x_{t+1} \in \mathcal{M}$, the on-manifold descent lemma S1.3, Equation (S3) applied at $(x, y) = (x_t, x_{t+1})$ yields

$$F(x_{t+1}) \leq F(x_t) + \langle \text{grad } F(x_t), x_{t+1} - x_t \rangle + \frac{L_{\mathcal{M}}}{2} \|x_{t+1} - x_t\|^2. \quad (\text{S9})$$

By Lemma 4.8 (Aggregation bias, proved in Section S3), for both RFedRL and RFedProj, we have the decomposition

$$x_{t+1} - x_t = -\alpha \frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) + \bar{b}_t + \xi_t$$

with $\mathbb{E}_t \|\bar{b}_t\| \leq O(\alpha^2)$, $\mathbb{E}_t \|\bar{b}_t\|^2 \leq O(\alpha^4)$ and $\mathbb{E}_t[\xi_t] = 0$, $\mathbb{E}_t \|\xi_t\|^2 = O\left(\frac{\alpha^2}{kB} \sigma_{\text{sg}}^2\right)$. Taking conditional expectation $\mathbb{E}_t[\cdot]$ in (S9) and using linearity,

$$\begin{aligned} \mathbb{E}_t[F(x_{t+1})] &\leq F(x_t) - \alpha \left\langle \text{grad } F(x_t), \mathbb{E}_t \left[\frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) \right] \right\rangle + \langle \text{grad } F(x_t), \mathbb{E}_t[\bar{b}_t] \rangle \\ &\quad + \underbrace{\langle \text{grad } F(x_t), \mathbb{E}_t[\xi_t] \rangle}_{=0} + \frac{L\mathcal{M}}{2} \mathbb{E}_t \|x_{t+1} - x_t\|^2, \end{aligned}$$

since $\text{grad } F(x_t)$ is $\mathcal{F}_t$ -measurable and $\mathbb{E}_t[\xi_t]$ has zero conditional mean. By sampling without replacement, $\mathbb{E}_t \left[\frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) \right] = \text{grad } F(x_t)$. For the inner product with $\bar{b}_t$, we use Cauchy-Schwarz and Young's inequality with parameter α :

$$\langle \text{grad } F(x_t), \mathbb{E}_t \bar{b}_t \rangle \leq \frac{\alpha}{2} \|\text{grad } F(x_t)\|^2 + \frac{1}{2\alpha} \mathbb{E}_t \|\bar{b}_t\|^2 \leq \frac{\alpha}{2} \|\text{grad } F(x_t)\|^2 + O(\alpha^3).$$

For the quadratic term, we have

$$\|x_{t+1} - x_t\|^2 \leq 2\alpha^2 \left\| \frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) \right\|^2 + 2\|\bar{b}_t + \xi_t\|^2$$

so that

$$\begin{aligned} \frac{L\mathcal{M}}{2} \mathbb{E}_t \|x_{t+1} - x_t\|^2 &\leq L\mathcal{M} \alpha^2 \mathbb{E}_t \left\| \frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) \right\|^2 + L\mathcal{M} \mathbb{E}_t \|\bar{b}_t + \xi_t\|^2 \\ &\leq L\mathcal{M} \alpha^2 \|\text{grad } F(x_t)\|^2 + O\left(\alpha^2 \frac{n-k}{k(n-1)} \sigma^2\right) \\ &\quad + 2L\mathcal{M} \mathbb{E}_t \|\bar{b}_t\|^2 + 2L\mathcal{M} \underbrace{\mathbb{E}_t \|\xi_t\|^2}_{=O\left(\frac{\alpha^2}{kB} \sigma_{\text{sg}}^2\right)} \end{aligned}$$

using

$$\mathbb{E}_t \left\| \frac{1}{k} \sum_{i \in S_t} \text{grad } f_i(x_t) \right\|^2 = \|\text{grad } F(x_t)\|^2 + \frac{n-k}{k(n-1)} \sigma^2.$$

Plugging everything into (S9) gives

$$\begin{aligned} \mathbb{E}_t[F(x_{t+1})] &\leq F(x_t) - \alpha \|\text{grad } F(x_t)\|^2 + \frac{\alpha}{2} \|\text{grad } F(x_t)\|^2 + L_{\mathcal{M}} \alpha^2 \|\text{grad } F(x_t)\|^2 \\ &\quad + O\left(\alpha^2 \left(\frac{n-k}{k(n-1)} \sigma^2 + \frac{\sigma_{\text{sg}}^2}{kB}\right)\right) + O(\alpha^3) + O(\alpha^4). \end{aligned}$$

Grouping the $\|\text{grad } F(x_t)\|^2$ terms gives $-(\alpha/2 - L_{\mathcal{M}} \alpha^2) \|\text{grad } F(x_t)\|^2$. By the stepsize condition $\alpha \leq 1/(4L_{\mathcal{M}})$, we have $\alpha/2 - L_{\mathcal{M}} \alpha^2 \geq \alpha/4$ hence

$$-(\alpha/2 - L_{\mathcal{M}} \alpha^2) \|\text{grad } F(x_t)\|^2 \leq -\frac{\alpha}{4} \|\text{grad } F(x_t)\|^2$$

S5. Proof of Lemma 4.10 (Stationarity from one-round descent)

Starting from Lemma 4.9 (One-round expected descent, proved in Section S4) and conditioning on $\mathcal{F}_t$,

$$\mathbb{E}_t[F(x_{t+1})] \leq F(x_t) - \frac{\alpha}{4} \|\text{grad } F(x_t)\|^2 + O\left(\alpha^2 \left(\frac{n-k}{k(n-1)} \sigma^2 + \frac{\sigma_{\text{sg}}^2}{kB}\right)\right) + O(\alpha^3) + O(\alpha^4).$$

Taking total expectation and summing over $t = 0, \dots, T-1$, by the tower property, we get

$$\begin{aligned} \mathbb{E}[F(x_T)] &\leq \mathbb{E}[F(x_0)] - \frac{\alpha}{4} \sum_{t=0}^{T-1} \mathbb{E} \|\text{grad } F(x_t)\|^2 \\ &\quad + T \left[O\left(\alpha^2 \left(\frac{n-k}{k(n-1)} \sigma^2 + \frac{\sigma_{\text{sg}}^2}{kB}\right)\right) + O(\alpha^3) + O(\alpha^4) \right]. \end{aligned}$$

Using $F(x_T) \geq F_{\star}$ a.s. and rearranging,

$$\begin{aligned} \frac{\alpha}{4} \sum_{t=0}^{T-1} \mathbb{E} \|\text{grad } F(x_t)\|^2 &\leq F(x_0) - F_{\star} \\ &\quad + T \left[O\left(\alpha^2 \left(\frac{n-k}{k(n-1)} \sigma^2 + \frac{\sigma_{\text{sg}}^2}{kB}\right)\right) + O(\alpha^3) + O(\alpha^4) \right]. \end{aligned}$$

Divide by T and then by $\alpha/4$ to obtain

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\text{grad } F(x_t)\|^2] \leq \frac{4(F(x_0) - F_\star)}{\alpha T} + O\left(\alpha \left(\frac{n-k}{k(n-1)}\sigma^2 + \frac{\sigma_{\text{sg}}^2}{kB}\right)\right) + O(\alpha^2) + O(\alpha^3). \quad (\text{S10})$$

S6. EEGNet architecture

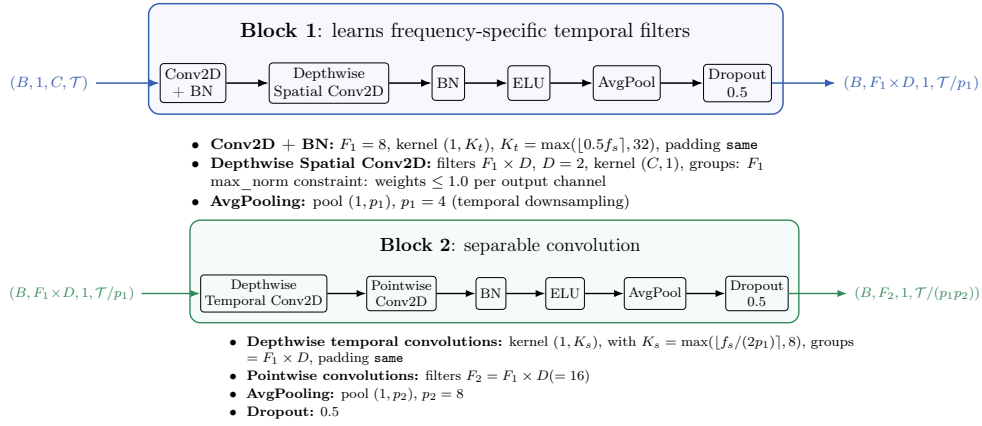


Figure S1: EEGNet architecture: (top) Block 1 for temporal and spatial feature extraction, (bottom) Block 2 for separable convolution refinement.

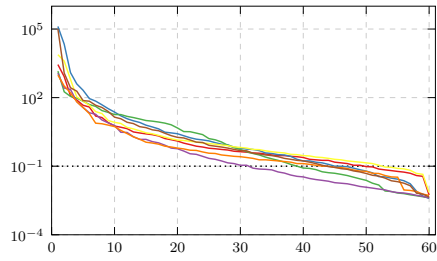
S7. SPDNet hyperparameter search

We detail here hyperparameter selection for SPDNet: eigenvalue spectral analysis to determine the regularization parameter ε , followed by hidden dimension sweeps to select the optimal architecture (see Figure S2).

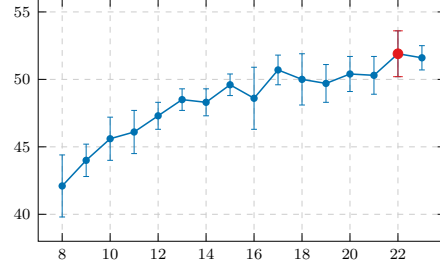
Spectral analysis. For each dataset, we compute sample covariance matrices from the EEG trials and examine their eigenvalue distributions to select the regularization parameter ε in the ReEig layer, which clamps small eigenvalues to ensure numerical stability during the matrix logarithm computation. The threshold must balance two concerns: too large a value discards meaningful signal, while too small a value leaves matrices ill-conditioned. We identify a threshold where eigenvalues decay below the noise floor, preserving discriminative information while ensuring positive definiteness. This analysis

uses the full dataset but does not introduce data leakage since it guides hyperparameter choices rather than model fitting. Each curve in the left column of Figure S2 shows the sorted eigenvalues of a representative sample covariance matrix per class; the dotted line indicates the selected ε .

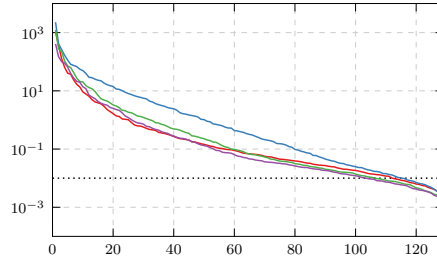
Hidden dimension selection. We perform a grid search over the hidden dimension h of the BiMap layer for the 1-block SPDNet architecture. For each hidden dimension, we train the model with 4 different random seeds and report the mean test F1 score with standard deviation. The regularization parameter ε is set based on the eigenvalue analysis above. A key motivation for this sweep is to identify SPDNet configurations that achieve competitive performance with significantly fewer parameters than EEGNet. Since the BiMap layer maps from $\mathcal{S}_{n_{\text{chan}}}^{++}$ to $\mathcal{S}_d^{++}$, reducing d dramatically decreases the model’s parameter count while leveraging the geometric structure of SPD manifolds. The results (see right column of Figure S2) demonstrate that relatively small hidden dimensions suffice to capture the discriminative information in the covariance matrices, yielding compact models suitable for federated learning scenarios where communication efficiency is critical.



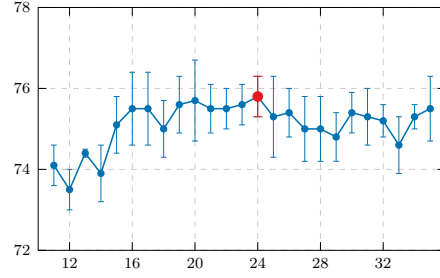
(a) Weibo2014, $\varepsilon = 0.1$



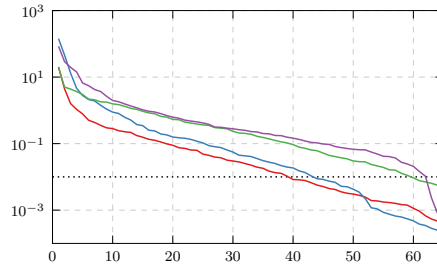
(b) Weibo2014: optimal $d = 22$, F1= 51.9%



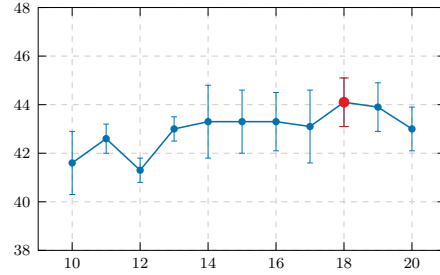
(c) Schirrmeister2017, $\varepsilon = 0.01$



(d) Schirrmeister2017: optimal $d = 24$, F1= 75.8%



(e) PhysionetMI (1/100 rescaled), $\varepsilon = 0.01$



(f) PhysionetMI: optimal $d = 18$, F1= 44.1%

— L. hand — R. hand — Hands — Feet — LH/RF — RH/LF — Rest
 ● Mean F1 ($\pm$ std) ● Optimal

Figure S2: Left: Eigenvalue spectra of mean covariance matrices per class (dotted line: regularization threshold). Right: Hidden dimension sweep for 1-block SPDNet showing test F1 (%).

S8. Additional figures

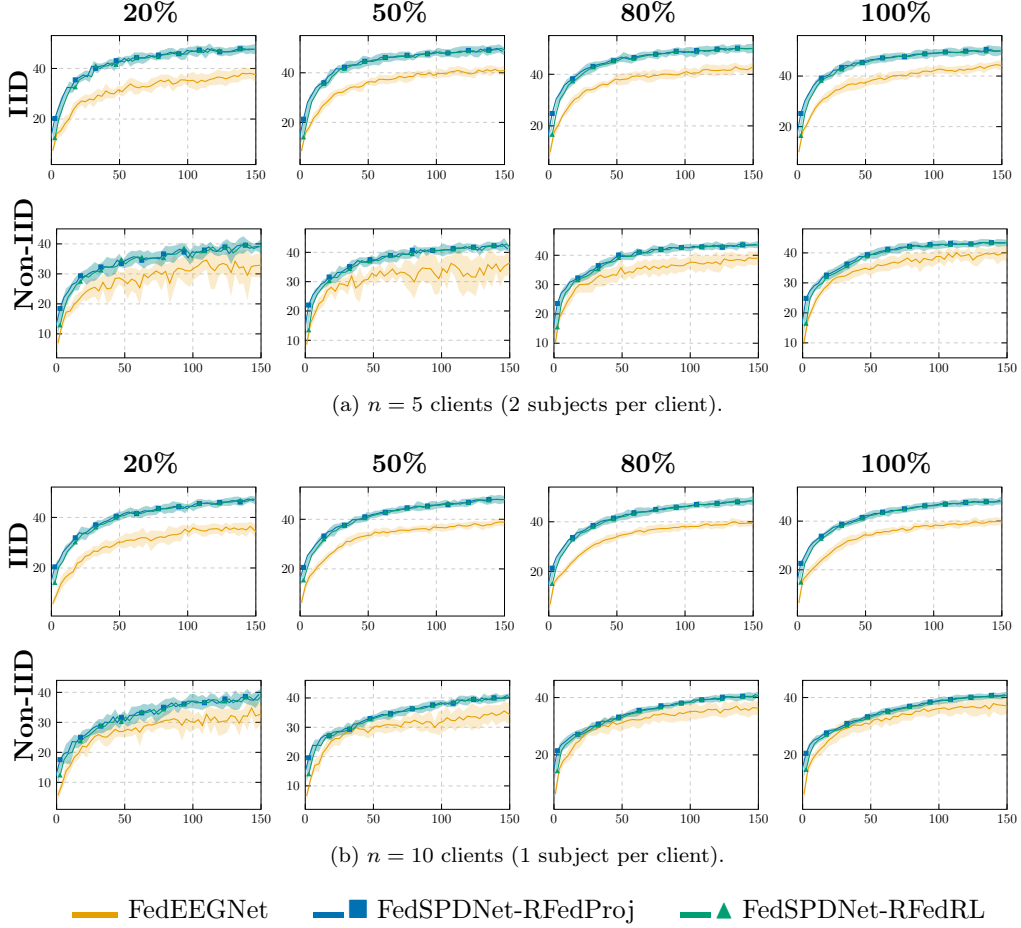
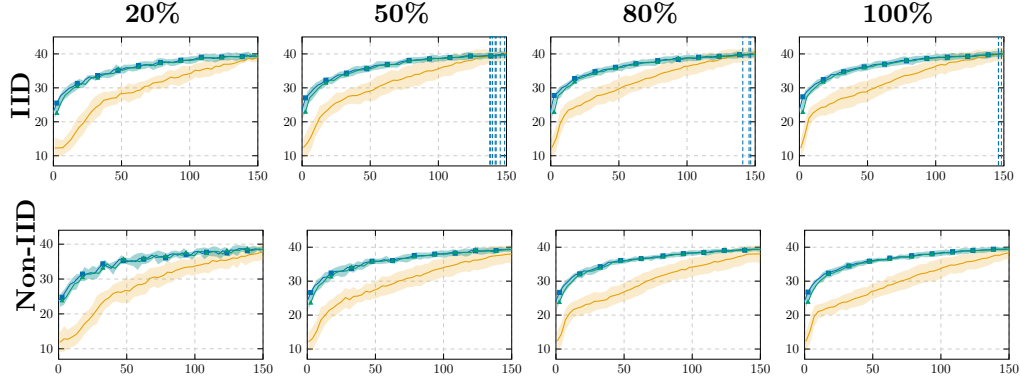
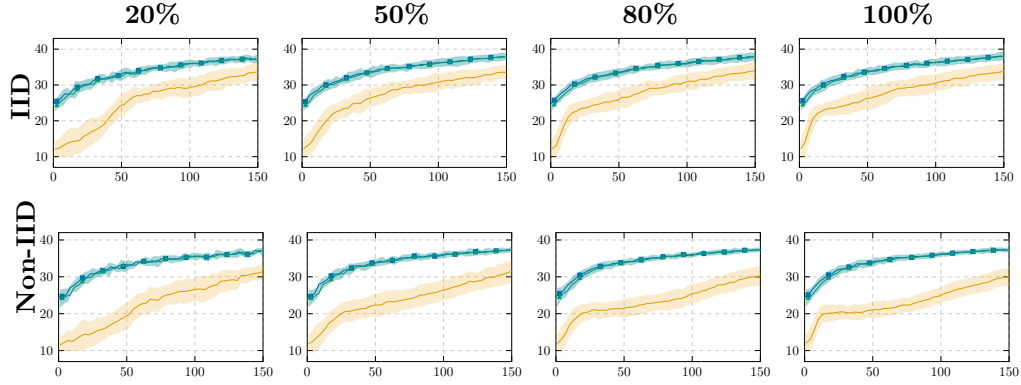


Figure S3: Convergence curves on Weibo2014 showing test F1 (%) vs. communication round under varying participation rates (columns) and data partitioning (rows). Shaded regions: ± 1 std over 10 seeds. RFedProj and RFedRL curves are slightly offset horizontally



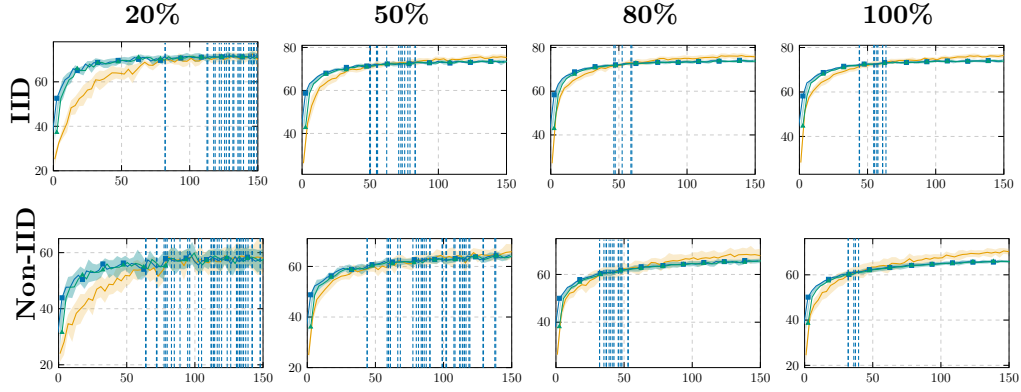
(a) $n = 53$ clients (2 subjects per client).



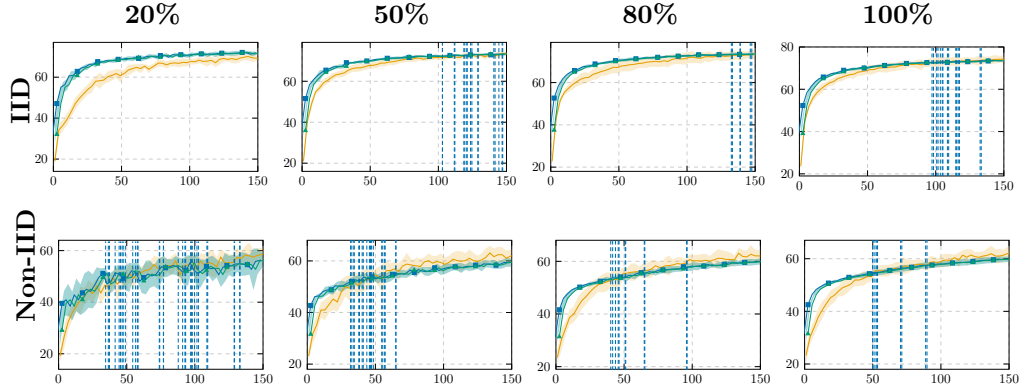
(b) $n = 106$ clients (1 subject per client).

— FedEEGNet
 —■— FedSPDNet-RFedProj
 —▲— FedSPDNet-RFedRL
- - - RFedProj crosses EEGNet

Figure S4: Convergence curves on PhysionetMI showing test F1 (%) vs. communication round under varying participation rates (columns) and data partitioning (rows). Shaded regions: ± 1 std over 10 seeds. Vertical dashed lines mark rounds at which FedSPDNet-RFedProj and FedEEGNet achieve equal test F1 performance. RFedProj and RFedRL curves are slightly offset horizontally



(a) $n = 7$ clients (2 subjects per client).



(b) $n = 14$ clients (1 subject per client).

— FedEEGNet
 —■ FedSPDNet-RFedProj
 —▲ FedSPDNet-RFedRL
- - - RFedProj crosses EEGNet

Figure S5: Convergence curves on Schirrmester2017 showing test F1 (%) vs. communication round under varying participation rates (columns) and data partitioning (rows). Shaded regions: ± 1 std over 10 seeds. Vertical dashed lines mark rounds at which FedSPDNet with RFedProj and FedEEGNet achieve equal test F1 performance. RFedProj and RFedRL curves are slightly offset horizontally.

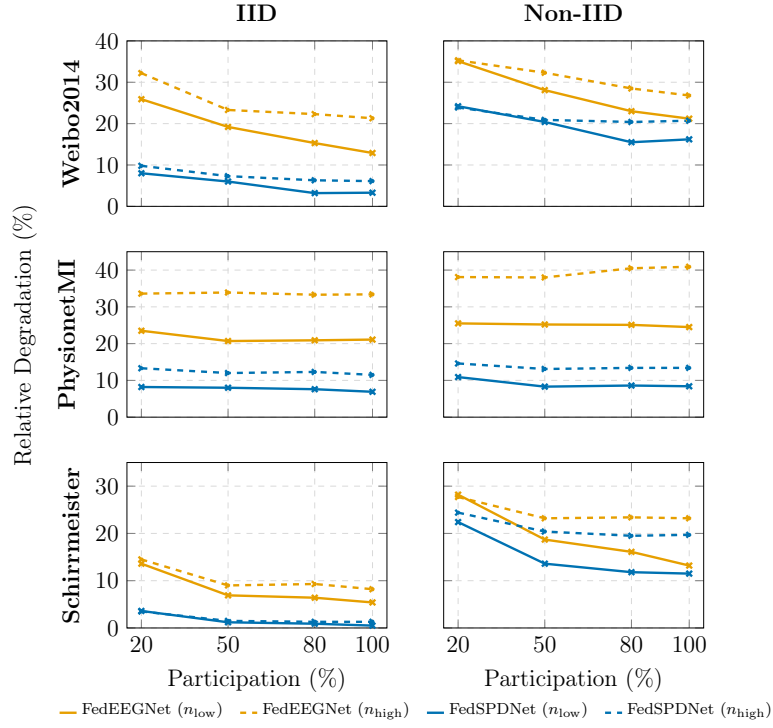


Figure S6: Relative performance degradation (%) from centralized baseline across datasets. Left column: IID partitioning. Right column: Non-IID (subject-based) partitioning. n_{low}/n_{high} : Weibo2014 (5/10), PhysionetMI (53/106), Schirrmeister2017 (7/14). Lower values indicate better robustness to federation.

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